# The Potential Outcome Framework

## PS200C, Causal Inferece, Spring 2023

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Reading: Chapers 1 and 2 from “What If" (Hernan & Robins) (google it)

# Purpose, Scope, and Examples

One kind of causal question—but not the only kind—regards the *effect* of some potential cause (e.g. an institution, intervention, policy, or event) on some outcome.

E.g., what is the effect of

- political institutions on corruption?
- effect of a drug or other therapy on a health outcome
- an ad on purchasing behavior
- peace-keeping missions on peace?
- church attendance on turnout?
- body cameras on police brutality?
- ...

# Two languages

Most academic work on causality employs one of two languages:

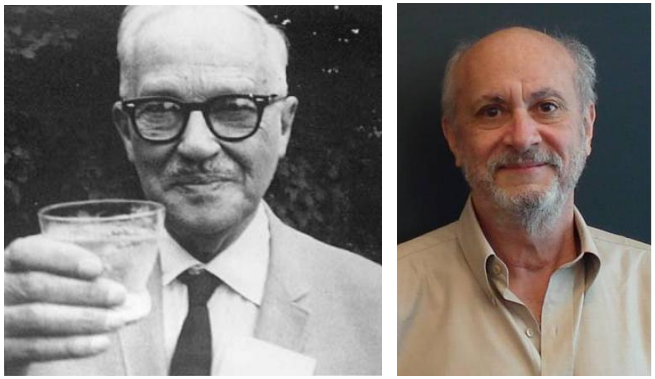


Figure: Neyman and Rubin

Figure: Judea Pearl

- Neyman-Rubin causal model, aka the Potential Outcomes Model (POM).
- Pearl associated with structural causal models (SCMs), graphs (directed acyclic graphs, DAGs)

We will use both extensively, but potential outcomes are more common in econometrics-informed areas. Let's start there.

# The Potential Outcomes Approach in a Nutshell

How would you know something had a causal effect? What does that mean?

The POM answer:

- Suppose unit $i$ takes a treatment and you measure the outcome. Let's call this $Y_{1i}$, or the treatment potential outcome
- You'd like to know how $Y_{1i}$ compares to the value of $Y$ unit $i$ *would have had, if it had not taken the treatment*. Call this $Y_{0i}$, or the non-treatment potential outcome
- Then unit $i$'s *treatment effect* is $\tau_i = Y_{1i} - Y_{0i}$.
- The trouble is you do not get to see both $Y_{1i}$ and $Y_{0i}$ for unit $i$. This is the fundamental problem of causal inference.
- Everything we do will be about filling in the "missing" potential outcome for each unit through various **assumptions**. That is it!

# Fundamental Quantities

- $Y_{1i}$ and $Y_{0i}$, the potential outcomes
- $D_i$, the treatment status of unit $i$
- Observed outcome, $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$
- $\tau_i = Y_{1i} - Y_{0i}$

For each, discuss whether they are AO (always observable) or SO (sometimes observable) or NO (never observable).

1. $Y_{1i}$ and $Y_{0i}$ (at same time)
2. $D_i$
3. $Y_i$
4. $\tau_i$

# Quantities of interest

Using these, we will soon be interested in things like:

- Average treatment effect (ATE): $\mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[\tau]$
- Average treatment effect among treated (ATET)

$$\mathbb{E}[\tau_i | D_i = 1] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1]$$

- Average treatment effect among treated (ATEC)

$$\mathbb{E}[\tau_i | D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 0]$$

# Warning: regression goggles

- You are used to thinking $D$ may be associated with $Y$, particularly if $D$ affects $Y$.
- But the potential outcomes are not the outcomes: they don't change when you change treatment, they say what would happen under a given treatment.
- In other words: the treatment doesn't change the potential outcomes, it changes which potential outcomes you see.
- The potential outcomes can even be independent of $D_i$

# Embodied assumptions

Just by writing down POs using notation $Y_i(d)$ we assert:

- consistency: $Y_i(d)$ is the $Y_i$ you'd see if $i$ received treatment $d$.
- no-interference: we've only asked if unit $i$ took the treatment, not unit $j$, so presumably $Y_i(D_i, D_j) = Y_i(D_i)$ regardless of the value of $D_j$.
- one-version-of-treatment – $d$ means the same thing all the time (I don't agree with this one, but you should know it's something people say due to Rubin 1980, see below)

# Embodied assumptions

Consistency is really definitional of POs...but worth naming because

- we will need to reference it later (as in $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_i|D_i = 1]$)
- to remind ourselves it can breakdown, e.g. due to different treatment versions or interference.

Non-interference + one-version are usually stated as part of Stable Unit Treatment Value assumption (SUTVA, Rubin 1980)

We can relax these in various ways, including:

- "treatment variation irrelevance" (VanderWeele 2009)
- expanding POs to index treatment of others in some limited way
- neglect

# Problematic POs

Just because you can write a PO doesn't mean you should.

- with a PO we imagine changing the treatment status of a unit
- if you cannot imagine the result, it may signal ill-definedness.
- one complication is that when you change the treatment status of unit $i$ it can affect a lot of things and this can confuse the causal quantity we are after

Example: what is the effect of leader gender on some outcome?

- for US presidents, this involves $Y_{trump}(female)$, i.e. "What would Donald Trump have done if he was female?'
- whenever you imagine "changing his sex", countless subsequent experiences would be different – did we want to include the effect of all those differences?
- And changing sex would greatly change the probability of being in the sample of presidents anyway.

In other cases, we commonly speak of some concept as if it means something specific, but its not clear what specifically one would mean to change about this as a treatment (e.g. obesity)

- A commonly proposed "rule" is that a treatment is only valid if you can imagine manipulating it, though it's not clear to me this is the right test.

# Train your brain

Let's try to automate some intuitions

Recall $D_i$ is a "switch" that determines whether you see $Y_{1i}$ or $Y_{0i}$

**Test 1**: Is $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_i|D_i = 1]$?

What licenses it?

# Train your brain

**Test 2**: Someone is assigning $D_i$ at random.

- What can we say about $\mathbb{E}[Y_{1i}|D_i = 1]$ compared to $\mathbb{E}[Y_{1i}|D_i = 0]$ and $\mathbb{E}[Y_{1i}]$?
- What would $\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$ tell us?

# Train your brain

**Test 3**: Suppose $Y_{1i} = Y_{0i}$ for all $i$. But now, someone (magically) sees $Y_{1i}$ for each unit and chooses $D_i = 1$ more often when $Y_{1i}$ is higher.

- How would the $Y_{1i}$ you observe compare to those of the rest of the population?
- So does $\mathbb{E}[Y_{1i}|D_i = 1] = \mathbb{E}[Y_{1i}|D_i = 0] = \mathbb{E}[Y_{1i}]$?
- How would the $Y_{0i}$ you see compare to that of the rest of the population?
- If you estimated $\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$, do you expect something near zero, positive, or negative?
- Can you say anything about expected direction of bias?

# Train your brain

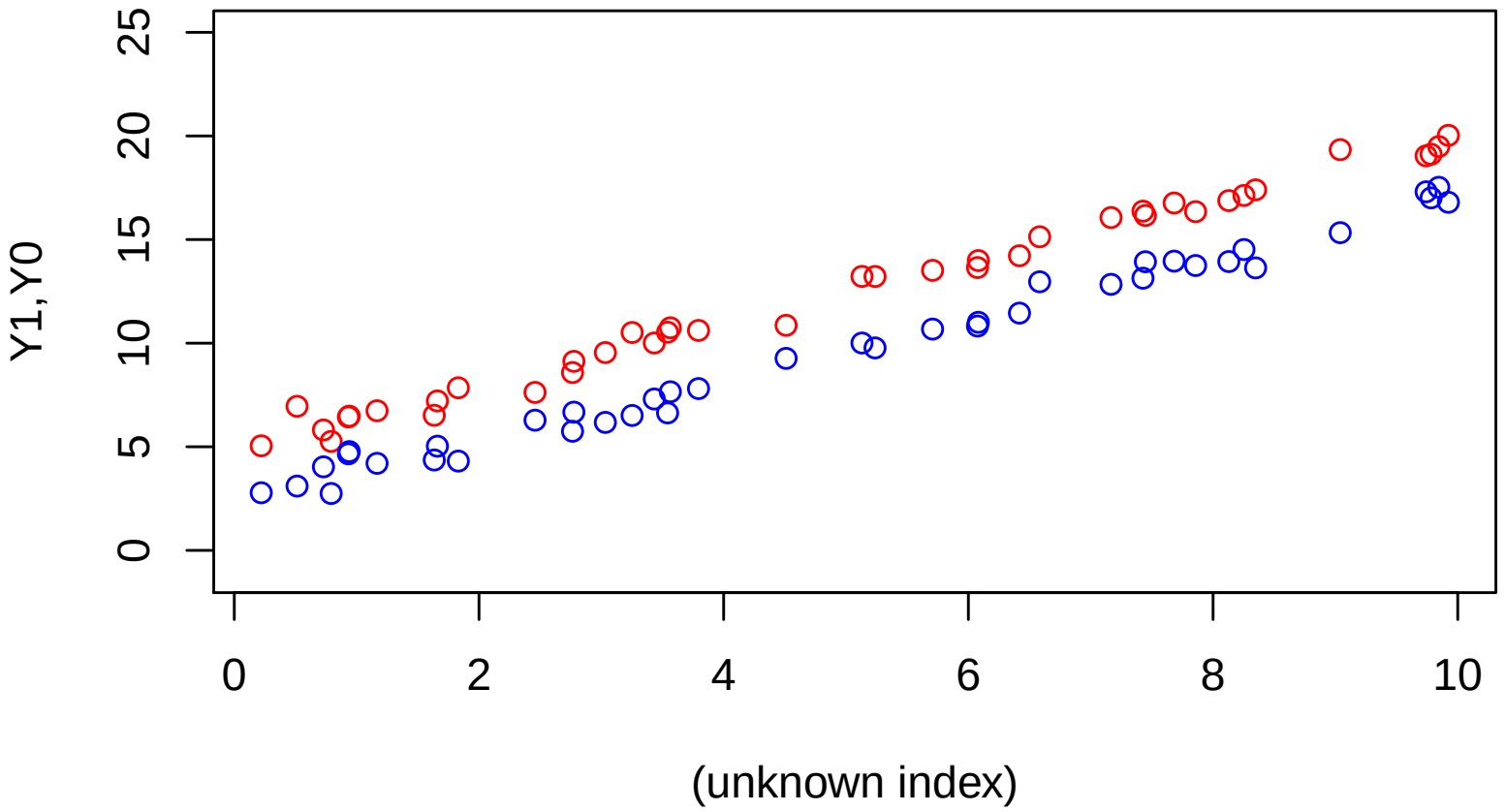
**Test 4**: Suppose you don't know anything about how $D_i$ is assigned w.r.t. $Y_{1i}$ and $Y_{0i}$.

- Can we say anything about how $\mathbb{E}[Y_1 i | D = 1]$ compares to $\mathbb{E}[Y_1 i]$?
- Can we say anything about how $\mathbb{E}[Y_0 i | D = 0]$ compares to $\mathbb{E}[Y_0 i]$?
- So can we say anything about how $\mathbb{E}[Y_i | D_i = 1] - \mathbb{E}[Y_i | D_i = 0]$ compares to $\mathbb{E}[Y_{1i} - Y_{0i}]$?

Which of these scenarios are we usually in with observational data?

# Visual Practice

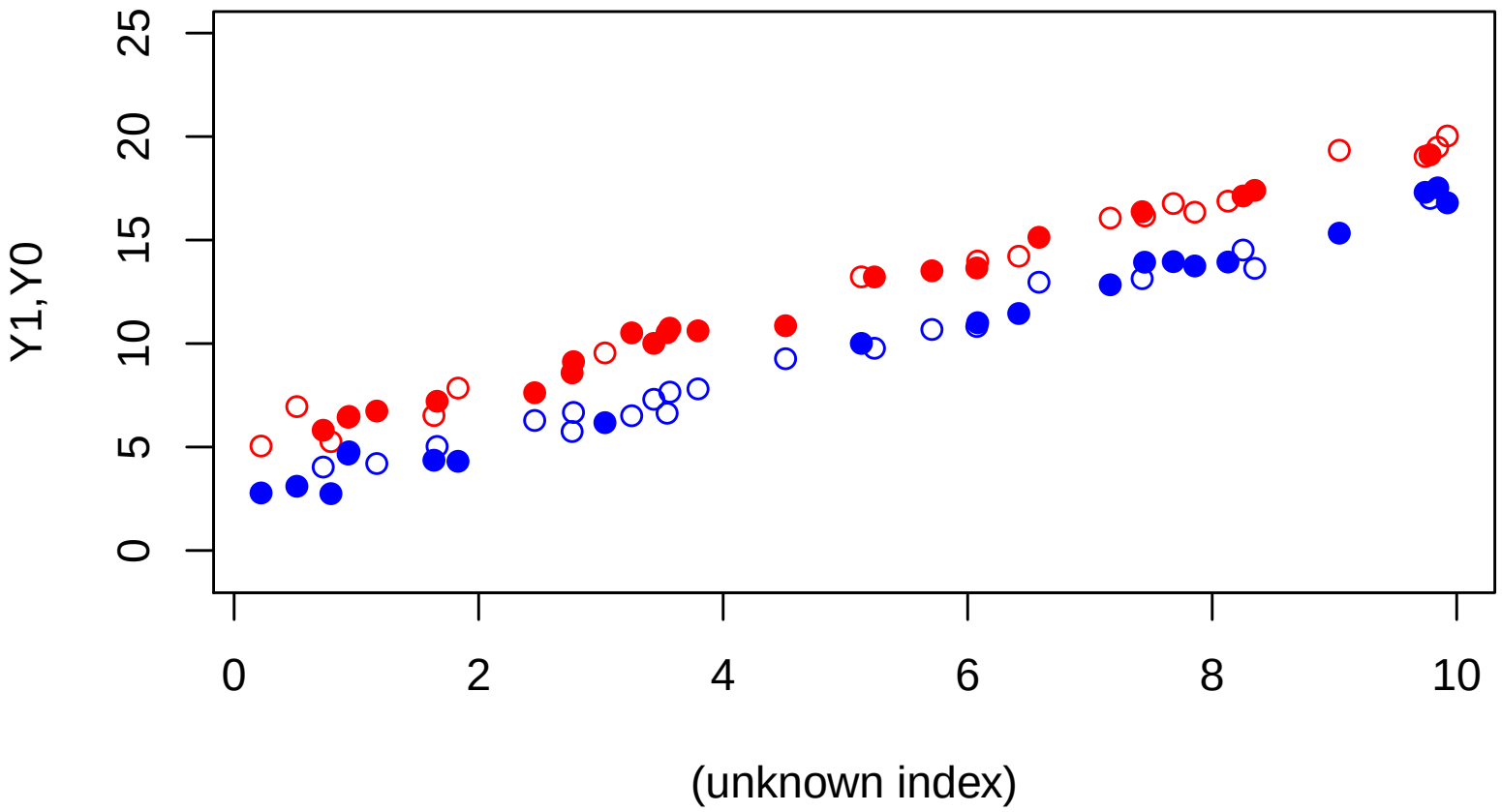
Let's looks at $Y_{1i}$ and $Y_{0i}$ alone first.




True $\mathbb{E}[Y_{1i} - Y_{0i}] = ATE = 3$

# Random Assignment of Treatment

Now suppose $D_i = 1$ is randomly assigned. We see only the filled dots:

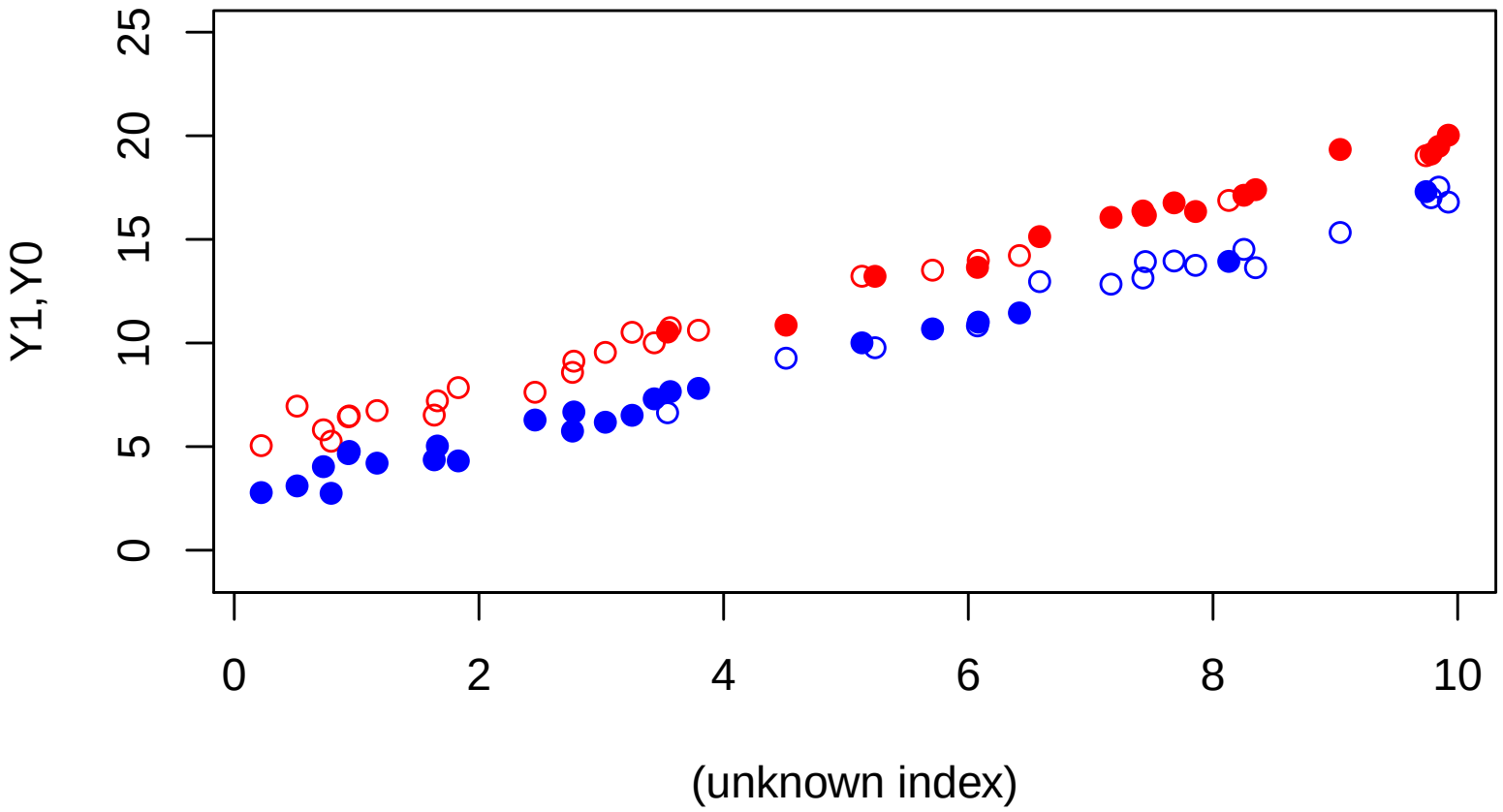



We get

$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = 3.01$$

# Non-Random Assignment of Treatment 1

Suppose $Pr(D_i = 1)$ increases to the right. Now we observe (filled dots):

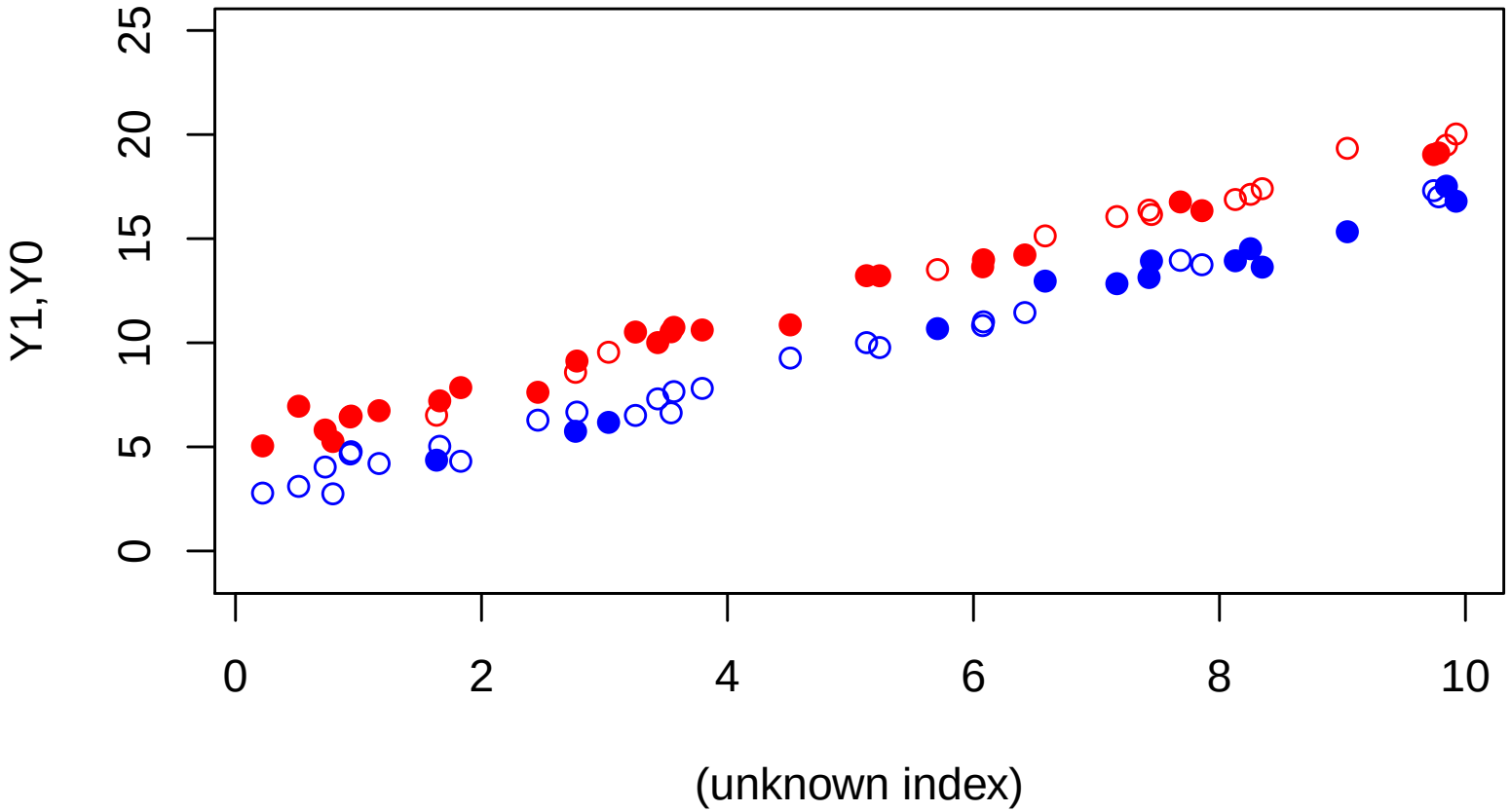



What difference in means do you expect?

$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = 9.97$$

# Non-Random Assignment of Treatment 2

Now suppose $Pr(D_i = 1)$ decreases to the right. We observe (filled dots):




What difference in means do you expect now?

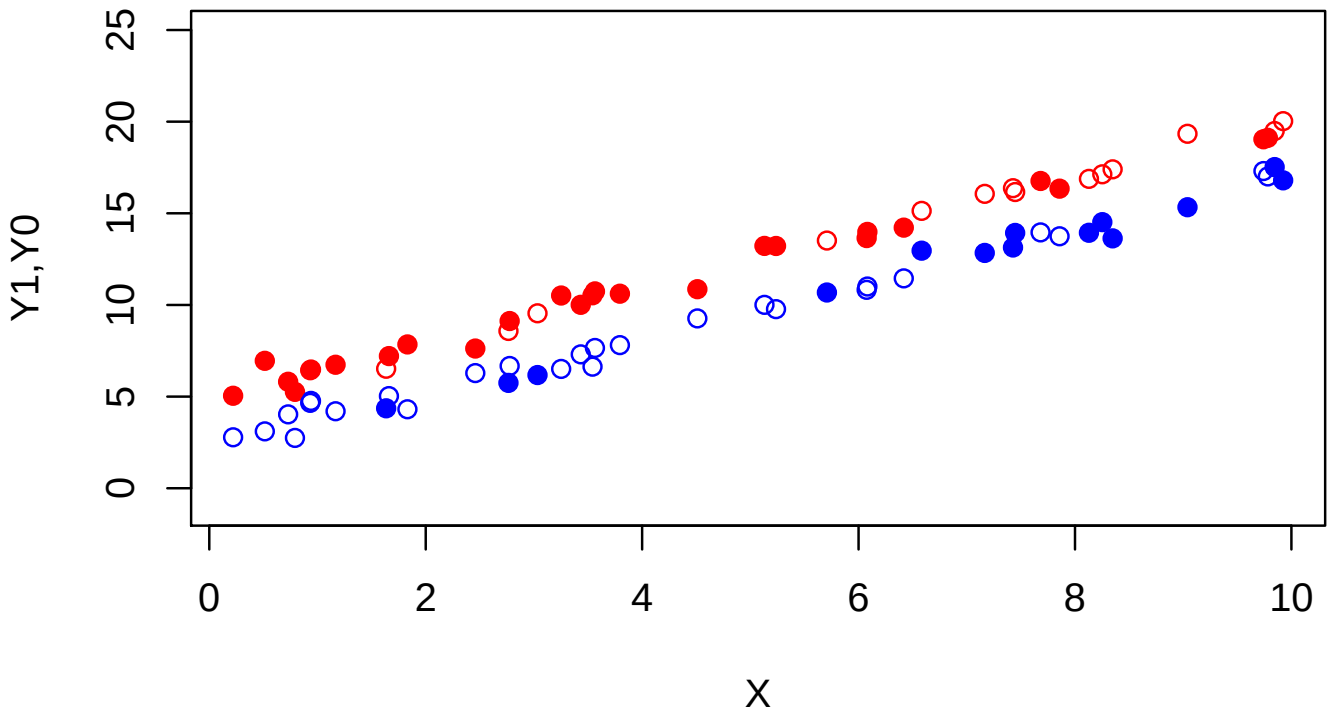
$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = -1.59$$

What can we do about this? Nothing, yet

# Preview (not for complete understanding)

Again, say $Pr(D_i = 1)$ decreases to the right

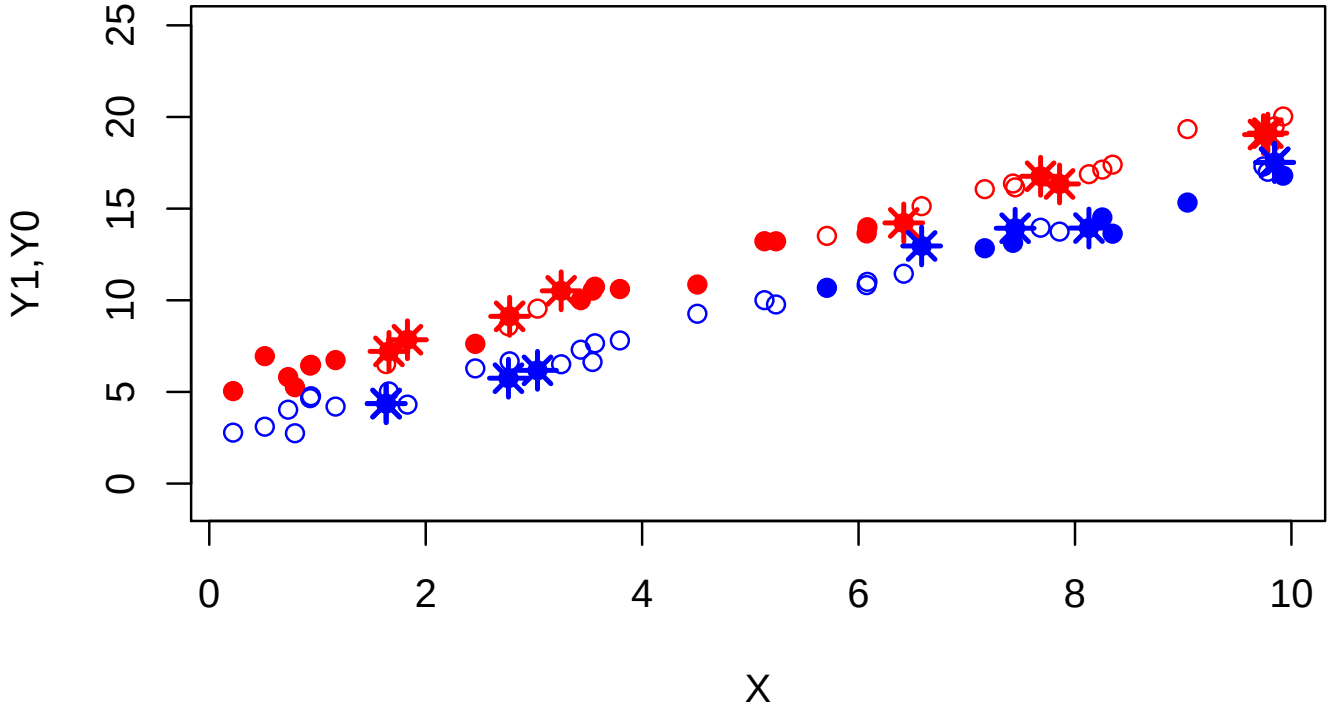
But now suppose $X$ is an observable, and $D$ is random conditional on $X$



# Preview of Things to Come

Again, say $Pr(D_i = 1)$ decreases to the right

But now suppose $X$ is an observable, and $D$ is random conditional on $X$




Having observed $X$ what if we compare treated to control only at starred points?

$$[\overline{Y}_i|D = 1] - [\overline{Y}_i|D = 0] = \widehat{ATE} = 2.63$$

# Repeat and refine: Quantities of usual interest

- Treatment effect, $\tau_i = Y_{1i} - Y_{0i}$
- Average Treatment Effect (ATE):

$$ATE = \mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[\tau_i] = \int \tau p(\tau) d\tau$$

- Average treatment effect on the treated (ATT):

$$ATT = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1] = \int \tau p(\tau | D = 1) d\tau$$

- Average treatment effect on the controls (ATC):

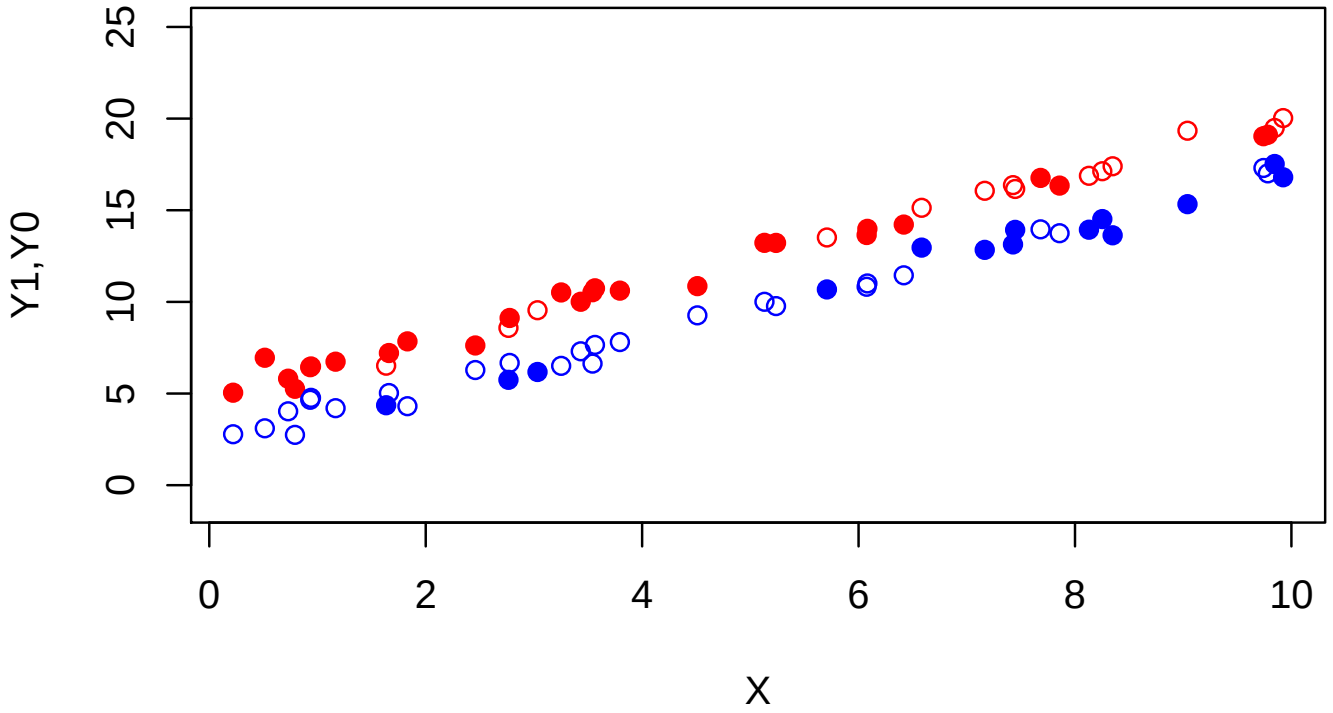
$$ATC = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 0] = \int \tau p(\tau | D = 0) d\tau$$

- Average treatment effect for sub-groups ($ATE(X)$):

$$ATE(X) = \mathbb{E}[Y_{1i} - Y_{0i} | X = x] = \int \tau p(\tau | X = x) d\tau$$

# Practice

Looking at this plot, describe the meaning of the $ATE$, $ATT$, $ATC$, & $ATE(X)$ graphically



# Back to SUTVA

Again, by writing POs as $\{Y_{1i}, Y_{0i}\}$ we imply we only care about treatment status of $i$, not $j$.

- And we defined the causal effect $\tau_i$ as $Y_{1i} - Y_{0i}$

Words you'll hear: Stable Unit Treatment Value Assumption (SUTVA), "no interference", or "individualized treatment response".

It is easy to break: disease, vaccination, conflict, education, anti-corruption monitoring, ...

Without this you get a proliferation of potential outcomes. E.g. with four units you could define:

$$\{Y_i(0,0,0,0), Y_i(0,0,0,1), ..., Y_i(1,1,1,1)\}$$

- how many potential outcomes for each unit, $i$?
- and how many causal effects for $i$?

# Potential outcomes with interference

**Definition (SUTVA)**

Unit $i$'s potential outcomes depend on $D_i$ and not on $D_{j \neq i}$

- e.g. $Y_3(0, 0, D_3, 0)$ is same as $Y_3(0, 1, D_3, 1)$, $Y_3(1, 1, D_3, 0)$,...
- So just write $Y_3(D_3)$
- Also limits the causal effects for each unit: simply $\tau_i = Y_i(1) - Y_i(0) = Y_{1i} - Y_{0i}$

This is an example of an **exclusion restriction**: we use outside information to rule out possibility of certain causal effects.

- the effect of you taking the treatment on my POs is excluded (0)
- traditional models (e.g. regression) do same: $Y_i$ depends on $X_i$ not $X_j$.

Causal inference in the presence of interference is an area of active research. e.g. sometimes "spillover" or "saturation" is deliberately varied

# More PO practice: the "science table"

Imagine a study population with 4 units:

| $i$ | $D_i$ | $Y_{1i}$ | $Y_{0i}$ | $\tau_i$ |
|---|---|---|---|---|
| 1 | 1 | 10 | 4 | 6 |
| 2 | 1 | 1 | 2 | -1 |
| 3 | 0 | 3 | 3 | 0 |
| 4 | 0 | 5 | 2 | 3 |

1. What is the ATE? $\mathbb{E}[Y_{1i} - Y_{0i}] = 1/4 \times (6 - 1 + 0 + 3) = 2$

(Note: average effect is positive, but not all $\tau_i$ are)

2. What are the ATT and ATC?

$$\mathbb{E}[Y_{1i} - Y_{0i}|D_i = 1] = .5(6 - 1) = 2.5$$

$$\mathbb{E}[Y_{1i} - Y_{0i}|D_i = 0] = .5(0 + 3) = 1.5$$

# Naive Comparison: Difference in Means

You saw earlier how simple comparison of observed outcomes can be misleading. Let's use POM to see this more rigorously.

The Difference in Means estimand is

$$\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]$$

POM helps us compare this to QOIs. One terrific decomposition:

$$\begin{aligned}\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] &= \mathbb{E}[Y_{1i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0] \\ &= \mathbb{E}[Y_{1i} - Y_{0i}|D_i = 1] + (\mathbb{E}[Y_{0i}|D_i = 1] - \mathbb{E}[Y_{0i}|D_i = 0])\end{aligned}$$

What are the terms in blue & green?

# Selection bias

$$\begin{aligned}\mathbb{E}[Y_i|D_i=1]-\mathbb{E}[Y_i|D_i=0] &= \mathbb{E}[Y_{1i}|D_i=1]-\mathbb{E}[Y_{0i}|D_i=0] \\ &= \mathbb{E}[Y_{1i}-Y_{0i}|D_i=1]+(\mathbb{E}[Y_{0i}|D_i=1]-\mathbb{E}[Y_{0i}|D_i=0]) \\ &= \text{ATT}+\text{selection bias}\end{aligned}$$

Example: $D =$ Church Attendance, $Y =$ Political Participation

- Church goers may differ from non-Church goers on a range of background characteristics (e.g. civic duty)
- Should you believe the naive DIM, $\mathbb{E}[Y_i|D_i=1]-\mathbb{E}[Y_i|D_i=0]$, reflects the causal effect of the church experience exclusively?
- Why not?...there could be some reason these groups are different "already", i.e. absent any effect of church itself: bias $= \mathbb{E}[Y_{0i}|D_i=1]-\mathbb{E}[Y_{0i}|D_i=0] \neq 0$

Example: Human rights treaties

- Maybe countries willing to sign a human rights treaty have better human rights (absent any effect of the treaty itself): bias $= \mathbb{E}[Y_{0i}|D_i=1]-\mathbb{E}[Y_{0i}|D_i=0] \neq 0$

# Assignment mechanism

We often start an analysis by thinking about how units are assigned to treatment.

We can classify many important examples (similar to the graphical examples we tried) by what we know of the assignment mechanism (for $D_i$):

- When we know it was random, we know those with $D = 1$ and $D = 0$ are "comparable", meaning same average POs.
- We previewed that if $D$ is random conditionally on observed $X$, we can manage.
- When we don't know anything about assignment, the data say nothing about the causal effects without importing some other assumption.

Those represent important assignment mechanisms with names: random assignment/ignorability, selection on observables/ conditional ignorability, and ignorance.

# Summing Up: The Neyman-Rubin / Potential Outcome Model

- Defines causality through counterfactual comparisons, defines various possible causal quantities of interest.
- In so doing, forces us to think about potential outcomes rather than just the observed outcome.
- There is no progress without estimating missing potential outcomes.
  - how do the potential outcomes we see differ from those we don't?
  - answer often comes through knowledge of the assignment mechanism, $p(D|Y_1, Y_0, X)$
- Writing POs commits us to embodied assumptions such as consistency and non-interference.

From here: random assignment, then random-conditional-on-observables.